

Missing Data Analysis - Assignment 1

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This assignment uses data.csv. The dataset contains a random sample of 500 individuals from a study investigating the association of Systolic Blood Pressure with LDL Cholesterol levels. The available variables are the Systolic Blood Pressure levels (SBP; in mmHg), the LDL Cholesterol levels (Chol; in mg/dl), the Age (in years) and the Gender of the participants.

Missing Data Overview

There are missing values in SBP and Chol but not in Age and Gender.

```
data <- read.table("data.csv", header = TRUE, sep = ",")
str(data)

## 'data.frame':    500 obs. of  4 variables:
## $ Age      : int  59 50 43 52 52 58 42 55 34 58 ...
## $ SBP      : int  114 NA NA NA 117 111 NA 117 NA 109 ...
## $ Chol     : int  NA NA NA 127 NA NA NA NA NA 136 ...
## $ Gender   : chr  "Male" "Male" "Female" "Female" ...
```

Task A

Sum of missing values for each variable

First, I'll count the total number of missing values (NA) for each variable across the dataset. Specifically, with `is.na()` function I'll create a matrix of logical values (TRUE for missing, FALSE for observed), and with `colSums()` sum these TRUE values by column. This will help me to identify which variables will later require imputation.

```
colSums(is.na(data)) #True for missing and False for observed values

##      Age      SBP      Chol Gender
##       0       241      399       0
```

We see that SBP has 241 missing values (48.2%) and Chol has 399 missing values (79.8%) out of 500 individuals.

Task B

B.1 Descriptive statistics

First I'll turn Gender into a factor of two levels (Male and Female), to make sure that R treats it correctly in statistical summary and later modeling.

```
data$Gender <- factor(data$Gender, levels = c("Male", "Female"))

# Overall dataset summary
summary(data)

##      Age      SBP      Chol      Gender
## Min.   :19.00  Min.   : 93.0  Min.   :104.0  Male   :270
## 1st Qu.:43.00  1st Qu.:114.0  1st Qu.:132.0  Female:230
## Median :51.00  Median :122.0  Median :145.0
## Mean   :50.15  Mean   :121.4  Mean   :143.4
## 3rd Qu.:58.00  3rd Qu.:129.0  3rd Qu.:153.0
## Max.   :80.00  Max.   :151.0  Max.   :180.0
##      NA's      NA's      NA's
##       0       241      399
```

To get a precise look at the continuous measures, I'll calculate the means for Age, SBP and Chol and their standard deviations with `colMeans()` and `sapply()` respectively. The `na.rm = TRUE` argument is used for SBP and Chol to remove their missing data. In other words, it is used to calculate the statistics only over observed cases and prevent NA as the output. Also, with `round()` the results will be rounded to two decimal places for clarity.

```
#Calculating individual Means
round(colMeans(data[, c("Age", "SBP", "Chol")], na.rm = TRUE), 2)

##      Age      SBP      Chol
## 50.15 121.35 143.40

# Calculating individual standard deviations
round(sapply(data[, c("Age", "SBP", "Chol")], sd, na.rm = TRUE), 2)

##      Age      SBP      Chol
## 10.68 11.76 15.46
```

And to get the frequency table and percentage distribution for the Gender variable.

```
# Frequency table
table(data$Gender)

##
##      Male Female
##      270      230

# Proportions / Percentages
prop.table(table(data$Gender)) * 100

##
##      Male Female
##       54       46
```

B.2 Pairwise Correlation Matrix with Listwise Deletion Method

In order to follow listwise deletion method to calculate the pairwise correlation matrix of SBP, Age and Chol, I will utilize the function `use = "complete.obs"`, with this each row where at least 1 of the three variables has NA will be discarded, and the matrix will be calculated based on the complete cases.

```
round(cor(data[, c("Age", "SBP", "Chol")], method = "pearson", use =
"complete.obs"), 2)

##      Age      SBP      Chol
## Age   1.00  0.37  0.48
## SBP   0.37  1.00  0.57
## Chol  0.48  0.57  1.00
```

B.3 Pairwise Correlation Matrix with Pairwise Deletion Method

With `use = "pairwise.complete.obs"`, correlations are calculated using all available data for each specific pair of variables. Since Age is fully observed, it is never the reason a row to be discarded. However, for pairs involving SBP or Chol, rows containing missing values for those variables will be excluded from those specific correlation calculations.

```
round(cor(data[, c("Age", "SBP", "Chol")], method = "pearson", use
="pairwise.complete.obs"), 2)
```

```
##      Age  SBP Chol
## Age   1.00 0.44 0.44
## SBP   0.44 1.00 0.57
## Chol  0.44 0.57 1.00
```

B.4 Differences in the two deletion methods

We can see differences in the calculation of the pairwise correlation matrix between listwise and pairwise deletion methods and they stem from varying sample sizes across pairs and shifting underlying subsets of data.

- Specifically, listwise deletion method calculates every correlation using the exact same subset of observations since it drops an entire row if any of their variables are missing (complete-case analysis). While this keeps the sample consistent across the matrix, it could reduce statistical power and introduce selection bias if the missingness isn't completely at random. Specifically it could remove entirely certain demographics (e.g. younger individuals), which could shift the relationship between the variables.
- On the other hand, pairwise deletion recalculates sample size for every pair, using all rows that have non-missing data for those two specific pair of variables, thus preserving a larger sample size (available-case analysis) and statistical power.

Task C

SBP dependence on Age and Missing Mechanism

I will look into the probability of missingness on Systolic Blood Pressure depending on the values of Age and the missingness mechanism by fitting a logistic regression model with missingness/observation of SBP as the dependent variable and Age as the (fully observed) covariate.

```
data$miss <- 1*is.na(data$SBP) # missingness indicator for Systolic Blood
Pressure (1=Missing, 0=Not Missing)
```

```
fitMiss <- glm(miss ~ Age, data = data, family = binomial) #logistic regression
model to see if Age predicts SBP missingness
```

```
summary(fitMiss) #p-value<0.05 and negative coef => MAR & Young People Less likely to answer
```

```
##
## Call:
## glm(formula = miss ~ Age, family = binomial, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.80126    0.51985   7.312 2.63e-13 ***
## Age         -0.07739    0.01022  -7.572 3.67e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 692.50  on 499  degrees of freedom
## Residual deviance: 623.07  on 498  degrees of freedom
## AIC: 627.07
##
## Number of Fisher Scoring iterations: 4
```

The logistic regression output shows that, Age is a highly significant predictor of whether SBP data is missing ($p\text{-value} < 0.001$). Therefore, the probability of missingness of SBP depends on the fully observed Age, the mechanism is at least Missing at Random (MAR) and not Missing Completely at Random (MCAR), but we cannot distinguish between MAR and MNAR. Furthermore, since the coefficient for Age is negative -0.077 it indicates that as age decreases, the log-odds (and thus the probability) of the data for SBP being missing increases. In other words, younger individuals are less likely to have their Systolic Blood Pressure recorded compared to older individuals.

Task D

Mean Imputation of SBP and Cholesterol

I'll perform mean imputation for both SBP and Chol. This replaces every missing value in a column by the overall average of the observed values in that same column.

```
#Mean value of SBP and Cholesterol before imputation
round(colMeans(data[, c("SBP", "Chol")], na.rm = TRUE), 2)

##    SBP    Chol
## 121.35 143.40

#SBP mean imputation

data$SBP_imp1 <- data$SBP #Copy of the original SBP column to preserve
```

```

observed data
data$SBP_imp1[is.na(data$SBP_imp1)] <- mean(data$SBP_imp1, na.rm = TRUE)
#Overwrite NAs with the mean of SBP
mean(data$SBP_imp1) #The mean is the same as in complete-case analysis

## [1] 121.3514

#Cholesterol mean imputation

data$Chol_imp <- data$Chol
data$Chol_imp[is.na(data$Chol_imp)] <- mean(data$Chol_imp, na.rm = TRUE)
mean(data$Chol_imp) #Identical mean as in complete-case analysis

## [1] 143.396

```

After imputation, the mean of the variables remain unchanged compared to the ones we obtain from the complete-case analysis.

Task E

E.1 Standard Regression Imputation

I'll posit a multiple linear regression with Systolic Blood Pressure as the dependent variable and Age, Gender as the independent variables based only on the complete cases. Then, I will fill the missing values in Systolic Blood Pressure by their predicted values from the fitted model and calculate the mean and standard error.

```

#Fit the multiple linear regression model
lm <- lm(SBP ~ Age + Gender, data = data)

#Generate predictions and imputed data
data$pred_values <- predict(lm, newdata = data) # generating predicted values for whole sample
data$SBP_imp2 <- data$pred_values # copy of the predicted values to preserve data
data$SBP_imp2[data$miss == 0] <- data$SBP[data$miss == 0] # replacing them with the observed value for rows with non-missing value on SBP

#Calculate mean
mean(data$SBP_imp2) # estimation based on the regression imputed data

## [1] 119.4151

#Calculate standard error
sd(data$SBP_imp2)/sqrt(nrow(data))

## [1] 0.4215148

```

The standard error (SE) here is 0.422. This is underestimated because the sample variance of the imputed dataset is artificially low.

E.2 Stochastic Regression Imputation

To solve the issue of artificially low variance caused by standard regression imputation, I will use Stochastic Regression Imputation. In stochastic regression imputation method, the predicted values are used as the imputed values, but adds a random value drawn from the normal distribution, as $\epsilon \sim N(0, \sigma^2)$, where σ^2 is the estimation of the variance of errors from the fitted model variance of the errors.

`sigma()` is used to extract the residual standard error from the linear model. Furthermore, For rows where SBP was missing (`miss == 1`), I will add random normal noise to the prediction. The noise has a mean = 0 and a standard deviation equal to the model's residual error `sd_errors`.

```
data$SBP_imp3 <- data$pred_values # again a copy of the predicted values

data$SBP_imp3[data$miss == 0] <- data$SBP[data$miss == 0] # replacing them
with the observed value for those with non-missing value on SBP
sd_errors <- sigma(lm) # estimation of standard deviation of errors from the
fitted model
set.seed(15) # making the results reproducible
data$SBP_imp3[data$miss == 1] <- data$SBP_imp3[data$miss == 1] +
rnorm(sum(data$miss == 1), 0, sd_errors) # adding a random error term to the
predicted values for those with missing on SBP

#Calculate mean
mean(data$SBP_imp3)

## [1] 119.3783

#Calculate standard error
sd(data$SBP_imp3)/sqrt(nrow(data))

## [1] 0.5427253
```

The SE 0.543 from the stochastic regression imputation method, is statistically more valid compared to the standard regression, because it accounts for the natural variance of the dataset.

E.3 Comparison of Standard Regression and Stochastic Regression imputation methods

Mean Comparison

The mean value for both methods is almost identical, specifically in the standard regression imputation we got 119.42 vs 119.38 in stochastic regression imputation, since both methods are unbiased estimators of the mean. This happens because the

missingness mechanism is Missing At Random (MAR) and the regression model is properly specified.

Standard Error Comparison

The standard regression imputation method reduces the dataset's variance, because all missing values are predicted exactly on the regression line and imputed values fall perfectly on the regression line (with zero residual variance). This results in smaller variance and the SE (0.422) of the deterministic method to be artificially smaller (underestimated) in value compared to the SE (0.542) of the stochastic imputation method. On the other hand, the dataset's variance in stochastic imputation is more realistic and the standard error more statistically valid.

General Comments

While the stochastic approach is a significant improvement in calculating the SE compared to standard regression imputation, the resulting SE 0.543 is still generally underestimated, because it is a single imputation method and the imputed data are treated as fixed/observed data. Thus resulting in ignoring the uncertainty inherent in the imputation process itself.

Extra: Validation with MICE

```
library("mice")
```

Another approach is using the `mice()` function from mice package with setting `method = norm.predict` and `method = norm.nob` for the Standard Regression Imputation and Stochastic Regression Imputation, respectively. This approach uses less calculations and command lines in order to execute compared with the previous method. Furthermore, the function `m = 1` limits the number of imputations (since we don't want multiple imputations) to just one and `maxit = 1` limits the number of iterations to one.

```
# Deterministic regression imputation
imp_det <- mice(data[, c("SBP", "Age", "Gender")], method = "norm.predict",
maxit = 1, m = 1) # Impute data

##
## iter imp variable
## 1 1 SBP

data_det <- complete(imp_det) # Creating the deterministic dataset with the
imputed values

# Calculating mean and standard error of the deterministic model
mean(data_det$SBP)

## [1] 119.4151

sd(data_det$SBP)/sqrt(nrow(data_det))
```



```
## [1] 0.4215148

# Stochastic regression imputation
imp_sto <- mice(data[, c("SBP", "Age", "Gender")], method = "norm.nob",
maxit = 1, m = 1) #Impute data

##
## iter imp variable
## 1 1 SBP

data_sto <- complete(imp_sto) # Creating the stochastic dataset with the
imputed values

# Calculating mean and standard error of the stochastic model
mean(data_sto$SBP)

## [1] 119.637

sd(data_sto$SBP)/sqrt(nrow(data_sto))

## [1] 0.5277048
```

As we can see we get the similar results for the mean and SE values. The slight differences in the values of SE (0.543 vs. 0.528) in the two ways of stochastic imputation methods stem from the fact that the `lm()` method and the `mice()` randomly drew different error values from the same distribution.